Fiscal Year:	FY 2012	**Task Last Updated:**	FY 02/01/2013
PI Name:	Bedford, Joel Ph.D.		
Project Title:	Dose-Rate Effects and Components of Systems Governing Variations in Susceptibility for Carcinogenic and Acute Radiation Risks following Gamma-Ray, Proton, or HZE Irradiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	970-491-7492
Organization Name:	Colorado State University		
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City:	Fort Collins	**State:**	CO
Zip Code:	80523	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
Start Date:	09/01/2007	**End Date:**	05/31/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date is now 5/31/2012 (previously 11/30/2011), per NSSC information (Ed., 11/29/2011) NOTE: End date is now 11/30/2011, per NSSC information (Ed., 5/31/2011) NOTE: Received NCE through 5/31/2011, per C. Guidry/JSC (08/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nagasawa , Hatsumi (Colorado State University) Peng, Yuanlin (Colorado State University) Weil, Michael (Colorado State University)		
Grant/Contract No.:	NNX07AP85G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The central hypothesis is that dose-rate effects in the 1 to 0.005 Gy/min range will be the same for protons vs. gamma-rays for chromosomal aberration induction. Second, levels of chromosomal radiosensitivity, and their dose-rate dependence will be similar for cells from individuals with genetic differences in radiosensitivity. Such a biomarker may help identify differences among individuals in a component contributing to their susceptibility to these effects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Knowledge of variation among individuals in their susceptibility to radiation carcinogenesis and the genetic determination of the extent of existing variation would be important information, not only of interest to individual astronauts who will be exposed to radiation during space missions, but would also be of interest to anyone who may be exposed to radiation in their occupation or in connection with activities or experiences where radiation exposure may be involved. To the extent that chromosomal aberrations are an important cellular endpoint of damage related to carcinogenesis this study would be directly relevant to that issue.
Task Progress:	The aims of this project were: 1) to compare dose-rate effects for gamma-rays vs. protons of 50MeV and 1 GeV vs 1GeV/n Fe ions for chromosomal aberration induction in cultured human cells. This would test assumptions regarding equivalence of gamma-ray and high energy protons with respect to RBEs and dose-rate effects of interest for astronauts. It would also test assumptions that for high LET radiations in general (especially with LETs of the order of 100 keV/u) there would a negligible dose rate effect . 2) to determine whether similar dose rate effects would apply for cells that have been reported to be mildly hypersensitive such as those derived ATM heterozygotes vs normal individuals. For low passage normal human fibroblasts and for MCF10A human mammary epithelial cells, the dose rate effect for induction of chromosomal aberrations seen for gamma-rays in the range of 70cGy/min (HDR) to 0.005cGy/min (LDR) was virtually identical for protons with energies of 1000MeV or 50MeV and corresponded to a dose rate reduction factor(DREF) of about 2.0 for all these radiations. There was a smaller, but still a significant dose rate effect for 1GeV/n Fe ions corresponding to a DREF of about 1.3 despite the fact that the LET for the Fe ions was approximately 150 keV/micrometer. For the MCF10A cells that we reported earlier, the responses (aberration induction, cell killing and gamma H2AX formation at low dose rates) in 2D or 3D culture were identical. We also found that indistinguishable dose rate effects were also seen for ATM +/- cells compared to ATM +/+ cells with the two cell strains that were derived from individuals who were siblings from the same family.
Bibliography Type:	Description: (Last Updated: 02/04/2013)
Articles in Peer-reviewed Journals	Peng Y, Nagasawa H, Warner C, Bedford JS. "Genetic susceptibility: radiation effects relevant to space travel." Health Physics. 2012 Nov;103(5):607-20. http://dx.doi.org/10.1097/HP.0b013e31826945b9 ; PubMed PMID: 23032891 , Nov-2012
Articles in Peer-reviewed Journals	Steffen LS, Bacher JW, Peng Y, Le PN, Ding LH, Genik PC, Ray FA, Bedford JS, Fallgren CM, Bailey SM, Ullrich RL, Weil MM, Story MD. "Molecular characterisation of murine acute myeloid leukaemia induced by 56Fe ion and 137Cs gamma ray irradiation." Mutagenesis. 2013 Jan;28(1):71-9. Epub 2012 Sep 17. http://dx.doi.org/10.1093/mutage/ges055 ; PubMed PMID: 22987027 , Jan-2013
Articles in Peer-reviewed Journals	Kato TA, Wilson PF, Nagasaw H, Peng Y, Weil MM, Little JB, Bedford JS. "Variations in radiosensitivity among individuals: a potential impact on risk assessment?" Health Physics. 2009 Nov;97(5):470-80. Review. http://dx.doi.org/10.1097/HP.0b013e3181b08eec ; PubMed PMID: 19820456 , Nov-2009
Articles in Peer-reviewed Journals	Nagasawa H, Little JB, Lin YF, So S, Kurimasa A, Peng Y, Brogan JR, Chen DJ, Bedford JS, Chen BP. "Differential role of DNA-PKcs phosphorylations and kinase activity in radiosensitivity and chromosomal instability." Radiat Res. 2011 Jan;175(1):83-9. http://dx.doi.org/10.1667/RR2092.1 ; PubMed PMID: 21175350 , Jan-2011
Articles in Peer-reviewed Journals	Li F, Huang Q, Chen J, Peng Y, Roop DR, Bedford JS, Li CY. "Apoptotic cells activate the "phoenix rising" pathway to promote wound healing and tissue regeneration." Sci Signal. 2010 Feb 23;3(110):ra13. http://dx.doi.org/10.1126/scisignal.2000634 ; PubMed PMID: 20179271 , Feb-2010
Articles in Peer-reviewed Journals	Li W, Li F, Huang Q, Shen J, Wolf F, He Y, Liu X, Hu YA, Bedford JS, Li CY. "Quantitative, noninvasive imaging of radiation-induced DNA double-strand breaks in vivo." Cancer Res. 2011 Jun 15;71(12):4130-7. http://dx.doi.org/10.1158/0008-5472.CAN-10-2540 ; PubMed PMID: 21527553 , Jun-2011
Articles in Peer-reviewed Journals	Huang Q, Li F, Liu X, Li W, Shi W, Liu FF, O'Sullivan B, He Z, Peng Y, Tan AC, Zhou L, Shen J, Han G, Wang XJ, Thorburn J, Thorburn A, Jimeno A, Raben D, Bedford JS, Li CY. "Caspase 3-mediated stimulation of tumor cell repopulation during cancer radiotherapy." Nat Med. 2011 Jul 3;17(7):860-6. http://dx.doi.org/10.1038/nm.2385 ; PubMed PMID: 21725296 , Jul-2011
Books/Book Chapters	Wilson PF, Bedford JS. "Radiobiological principles." in "Textbook of Radiation Oncology. 3rd edition." Ed. R. Hoppe, T.L. Phillips, S.A. Leibel. Philadelphia : Elsevier/Saunders, 2010. Chapter 1., Sep-2010